



CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS

DATE: 24/07/2026

EXPERIMENT 9 -- Query with Joins – EquiJoin, InnerJoin, OuterJoin

AIM:

To perform **Equi Join, Inner Join, Left Join, and Right Join** on the student_cgpa and department tables using MySQL Workbench.

PROCEDURE:

1. Open MySQL Workbench and connect to the database.
2. Create the student_cgpa and department tables.
3. Insert records into both tables.
4. Execute Equi Join, Inner Join, Left Join, and Right Join queries.
5. Verify the results displayed in the Result Grid.
6. Conclude the experiment.

PROGRAM:

1. CREATE TABLE

```
CREATE TABLE student_cgpa (  
    reg_no VARCHAR(15) PRIMARY KEY,  
    student_name VARCHAR(50) NOT NULL,  
    department VARCHAR(20),  
    no_of_subs_pass INT,  
    no_of_subs_fail INT,  
    cgpa DECIMAL(3,2)  
);
```

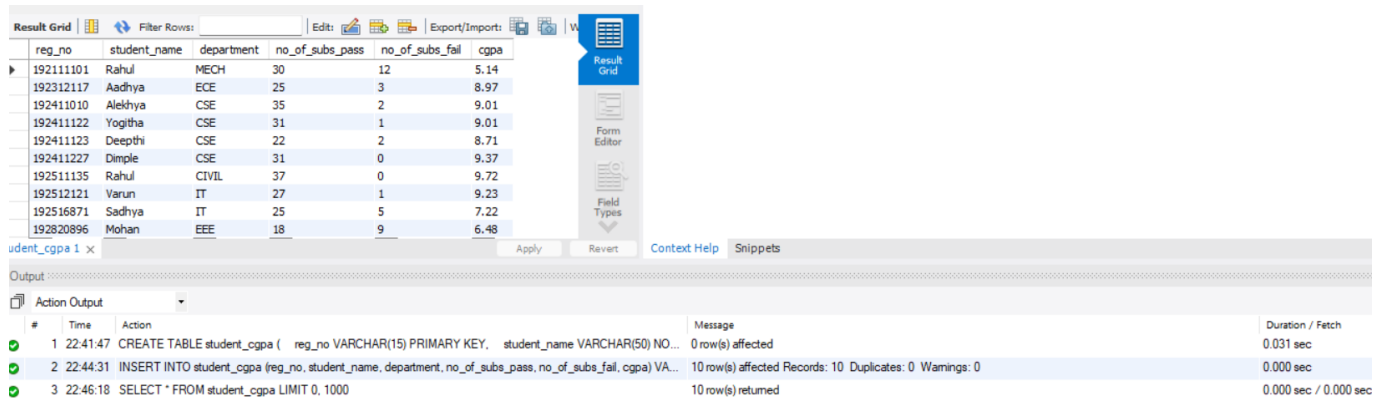
Then Insert Values:

```
INSERT INTO student_cgpa  
(reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa)  
VALUES  
( '192411123', 'Deepthi', 'CSE', 22, 2, 8.71),  
( '192512121', 'Varun', 'IT', 27, 1, 9.23),  
( '192312117', 'Aadhya', 'ECE', 25, 3, 8.97),  
( '192411227', 'Dimple', 'CSE', 31, 0, 9.37),  
( '192820896', 'Mohan', 'EEE', 18, 9, 6.48),  
( '192411122', 'Yogitha', 'CSE', 31, 1, 9.01),  
( '192516871', 'Sadhya', 'IT', 25, 5, 7.22),
```

```
(192111101,'Rahul','MECH',30,12,5.14),
(192411010,'Alekhyia','CSE',35,2,9.01),
(192511135,'Rahul','CIVIL',37,0,9.72);
```

```
SELECT * FROM student_cgpa;
```

OUTPUT:



The screenshot shows a database management tool interface. The top part displays a 'Result Grid' with the following data:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192111101	Rahul	MECH	30	12	5.14
192312117	Aadhya	ECE	25	3	8.97
192411010	Alekhyia	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411123	Deepthi	CSE	22	2	8.71
192411227	Dimple	CSE	31	0	9.37
192511135	Rahul	CIVIL	37	0	9.72
192512121	Varun	IT	27	1	9.23
192516871	Sadhya	IT	25	5	7.22
192820896	Mohan	EEE	18	9	6.48

The bottom part shows the 'Action Output' window with the following log:

#	Time	Action	Message	Duration / Fetch
1	22:41:47	CREATE TABLE student_cgpa (reg_no VARCHAR(15) PRIMARY KEY, student_name VARCHAR(50) NO...	0 row(s) affected	0.031 sec
2	22:44:31	INSERT INTO student_cgpa (reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa) VA...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	22:46:18	SELECT * FROM student_cgpa LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

2. CREATE DEPARTMENT TABLE

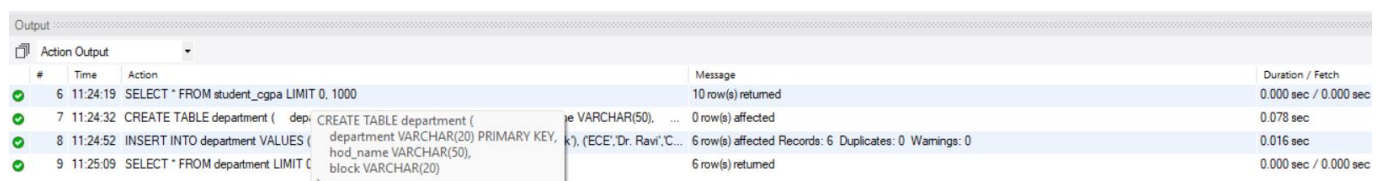
```
CREATE TABLE department (
department VARCHAR(20) PRIMARY KEY,
hod_name VARCHAR(50),
block VARCHAR(20)
);
```

Insert Values

```
INSERT INTO department VALUES
('CSE','Dr. Kumar','A Block'),
('IT','Dr. Meena','B Block'),
('ECE','Dr. Ravi','C Block'),
('EEE','Dr. Priya','D Block'),
('MECH','Dr. Suresh','E Block'),
('CIVIL','Dr. Ramesh','F Block');
```

```
SELECT * FROM department;
```

OUTPUT:



The screenshot shows the 'Action Output' window with the following log:

#	Time	Action	Message	Duration / Fetch
6	11:24:19	SELECT * FROM student_cgpa LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
7	11:24:32	CREATE TABLE department (department VARCHAR(20) PRIMARY KEY, hod_name VARCHAR(50), block VARCHAR(20)	0 row(s) affected	0.078 sec
8	11:24:52	INSERT INTO department VALUES (department VARCHAR(20) PRIMARY KEY, hod_name VARCHAR(50), block VARCHAR(20)	6 row(s) affected Records: 6 Duplicates: 0 Warnings: 0	0.016 sec
9	11:25:09	SELECT * FROM department LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec

Result Grid			
	department	hod_name	block
▶	CIVIL	Dr. Ramesh	F Block
	CSE	Dr. Kumar	A Block
	ECE	Dr. Ravi	C Block
	EEE	Dr. Priya	D Block
	IT	Dr. Meena	B Block
	MECH	Dr. Suresh	E Block
•	NULL	NULL	NULL

1. Equi Join

List the student name, department, and HOD name.

```
SELECT s.student_name, s.department, d.hod_name
FROM student_cgpa s, department d
WHERE s.department = d.department;
```

OUTPUT:

Result Grid			
	student_name	department	hod_name
▶	Rahul	MECH	Dr. Suresh
	Aadhya	ECE	Dr. Ravi
	Alekhya	CSE	Dr. Kumar
	Yogitha	CSE	Dr. Kumar
	Deepthi	CSE	Dr. Kumar
	Dimple	CSE	Dr. Kumar
	Rahul	CIVIL	Dr. Ramesh
	Varun	IT	Dr. Meena
	Sadhya	IT	Dr. Meena
	Mohan	EEE	Dr. Priya

Result 5 ×

2. Inner Join

Display the registration number, student name, department, and HOD name.

```
SELECT s.reg_no, s.student_name, s.department, d.hod_name
FROM student_cgpa s
INNER JOIN department d
ON s.department = d.department;
```

OUTPUT:

Result Grid					Filter Rows:	Export:
	reg_no	student_name	department	hod_name		
▶	192111101	Rahul	MECH	Dr. Suresh		
	192312117	Aadhya	ECE	Dr. Ravi	Dr. Sur	
	192411010	Alekhya	CSE	Dr. Kumar		
	192411122	Yogitha	CSE	Dr. Kumar		
	192411123	Deepthi	CSE	Dr. Kumar		
	192411227	Dimple	CSE	Dr. Kumar		
	192511135	Rahul	CIVIL	Dr. Ramesh		
	192512121	Varun	IT	Dr. Meena		
	192516871	Sadhya	IT	Dr. Meena		
	192820896	Mohan	EEE	Dr. Priya		

3. Left Outer Join

Display all students along with their HOD names (if a matching department exists).

```
SELECT s.student_name, s.department, d.hod_name
FROM student_cgpa s
LEFT OUTER JOIN department d
ON s.department = d.department;
```

OUTPUT:

Result Grid				Filter Rows:
	student_name	department	hod_name	
▶	Rahul	MECH	Dr. Suresh	
	Aadhya	ECE	Dr. Ravi	
	Alekhya	CSE	Dr. Kumar	
	Yogitha	CSE	Dr. Kumar	
	Deepthi	CSE	Dr. Kumar	
	Dimple	CSE	Dr. Kumar	
	Rahul	CIVIL	Dr. Ramesh	
	Varun	IT	Dr. Meena	
	Sadhya	IT	Dr. Meena	
	Mohan	EEE	Dr. Priya	

RESULT:

Thus, the JOIN operations (Equi Join, Inner Join, and Left Outer Join) were successfully performed on the student_cgpa and department tables using MySQL Workbench, and the required records were retrieved successfully.